Probability and Random Processes

Gaussian random vectors.
Exercises

1. Let S and N be independent random variables such that $S \sim \mathsf{N}(m_S, \sigma_S^2)$ and $N \sim \mathsf{N}(0, \sigma_N^2)$. If $Z = S + N$, find the joint characteristic function of Z and S.

2. Prove that the standard normal distribution $\mathsf{N}(0,1)$ has all its moments of odd order equal to 0 while the moments of even order are given by

$$\mu_{2j} = \frac{(2j)!}{2^j(j!)} \quad j = 0, 1, 2, \ldots$$

3. Let $X \sim \mathsf{N}(0,1)$ be a standard normal random variable.

(a) Prove that the characteristic function of X^2 is $(1 - 2i\omega)^{-1/2}$. [*Hint:* If $a \in \mathbb{R}$, then $\int_{-\infty}^{\infty} e^{(-1+ia)t^2} dt = \sqrt{\pi}/\sqrt{1 - ia}$.]

(b) Find the characteristic function of the random variable $X_1^2 + \cdots + X_n^2$, where the variables X_i, $1 \leqslant i \leqslant n$, are all standard normal and independent.

4. Let X_1 and X_2 be Gaussian random variables with mean 0 and covariance matrix $\begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$. If $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$, find the characteristic function of (Y_1, Y_2) and the marginal density of Y_1.

5. Find the covariance matrix of the jointly Gaussian random variables X and Y if we know that $(2X - Y)/3$ and $(X + Y)/3$ are $\mathsf{N}(0,1)$ independent random variables.

6. Let X_1, X_2, and X_3 be jointly Gaussian random variables with mean 1 and a covariance matrix

$$\begin{pmatrix} 1 & 1/2 & -1/2 \\ 1/2 & 1 & 1/2 \\ -1/2 & 1/2 & 1 \end{pmatrix}.$$

If $Y_1 = X_1 + X_2 - X_3$ and $Y_2 = X_1 - 2X_3$, find the covariance matrix of (Y_1, Y_2), the marginal densities of Y_1 and Y_2, and $\mathbb{P}(X_3 < X_1 + X_2)$.

7. Let X and Y be continuous random variables and let $Z = aX + bY$ where $a, b \in \mathbb{R}$.

(a) Prove that if the density f_Z is known for all $a, b \in \mathbb{R}$, then the joint density f_{XY} is univocally determined.

(b) Prove that if the random variable Z is Gaussian for all $a, b \in \mathbb{R}$, then X and Y are jointly Gaussian.

8. Let Y_1 and Y_2 be jointly Gaussian random variables such that $\mathbb{E}(Y_1) = 1$, $\mathbb{E}(Y_2) = -1$, $\mathrm{Var}(Y_1) = 4$, $\mathrm{Var}(Y_2) = 1$ and $\rho = 1/2$. Let N be also Gaussian with $\mathbb{E}(N) = 0$, $\mathrm{Var}(N) = 2$ and independent of (Y_1, Y_2). If $X = Y_1 - Y_2 + N$, prove that (X, Y_1) is a Gaussian vector and compute its parameters.

9. (a) Let X and Y be random variables with characteristic functions M_X and M_Y, respectively, and a joint characteristic function M_{XY}. A necessary and sufficient condition for the independence of X and Y is that $M_{XY}(\omega_1, \omega_2) = M_X(\omega_1) M_Y(\omega_2)$. Prove the necessity of this condition.

(b) Let X and Y be independent Gaussian $\mathsf{N}(0,1)$ random variables. Use characteristic functions to show that $S = X + Y$ and $R = X - Y$ are independent.

10. The independent r.v. X_i ($i \geqslant 1$) are all Gaussian with mean 1 and variance 2. Find $\mathbb{P}(X_{n+1} > X_1 + X_2 + \cdots + X_n)$. Express the result in terms of $F_{\mathsf{N}(0,1)}$.

11. Let X, Y, Z be jointly Gaussian random variables such that $m_X = m_Y = 0$, $m_Z = -1$, $\sigma_X^2 = \sigma_Y^2 = \sigma_Z^2 = 1$. Moreover, we know that X and Y are uncorrelated, that the correlation coefficient of X and Z is $1/2$ and that the correlation coefficient of Y and Z is $3/4$. Calculate the variance of $X + 2Y - 3Z$.

12. Let X be a Gaussian random variable with expectation vector $\boldsymbol{m} = (1, -1, 2)^t$ and covariance matrix

$$\boldsymbol{K} = \begin{pmatrix} 4 & 0 & -1 \\ 0 & 5 & 0 \\ -1 & 0 & 2 \end{pmatrix}$$

Which of the following random variables are independent: X_1 and X_2; X_1 and X_3; X_2 and X_3; X_1 and $X_1 - 3X_2 + 4X_3$?

13. Let $\boldsymbol{X} = (X_1, X_2, X_3)^t$ have a three-dimensional normal distribution.

(a) Show that if X_1 and $X_2 + X_3$ are uncorrelated, X_2 and $X_1 + X_3$ are uncorrelated, and X_3 and $X_1 + X_2$ are uncorrelated, then X_1, X_2, and X_3 are independent.

(b) Give and example for which X_1 and $X_2 + X_3$ are uncorrelated, X_2 and $X_1 + X_3$ are uncorrelated, and X_3 and $X_1 - X_2$ are uncorrelated, but X_1, X_2, and X_3 are not independent.

14. Let $(X_1, X_2, X_3)^t$ be a Gaussian random vector with expectation $(0, 0, 0)^t$ and covariance matrix $\boldsymbol{C}$.

(a) Let $\boldsymbol{C}'$ be the sub-matrix of $\boldsymbol{C}$ corresponding to X_2 and X_3. Prove that if $\det(\boldsymbol{C}) > 0$ and $\det(\boldsymbol{C}') > 0$, then the conditional distribution of X_1 given that $X_2 = X_3 = 0$ is Gaussian with mean 0 and variance $\det(\boldsymbol{C})/\det(\boldsymbol{C}')$. [*Hint*: consider conditional densities.]

(b) Let

$$\boldsymbol{C} = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 3 & 0 \\ -1 & 0 & 5 \end{pmatrix}.$$

Find the conditional distribution of X_1 given that $X_1 = X_2$ and $X_1 + X_2 + X_3 = 0$.

15. Let X be a Gaussian $\mathsf{N}(0, 1)$ random variable. Let Y given $X = x$ be Gaussian $\mathsf{N}(x, 1)$.

(a) Prove that (X, Y) is a Gaussian random vector and find its expectation vector and covariance matrix.

(b) Find the density of the variable Y.

16. Let X be a continuous random variable with a probability density function $f_X(x)$, such that $f_X(-x) = f_X(x)$ for all $x \in \mathbb{R}$. Let S be a discrete random variable that is independent of X, and such that S takes the values -1 and 1 with probabilities $1 - p$ and p, respectively, where $0 < p < 1$. Finally, define $Y = SX$.

(a) Show that the characteristic function of X, $M_X(\omega)$, satisfies the property $M_X(-\omega) = M_X(\omega)$.

(b) Use characteristic functions to prove that X and Y are identically distributed.

(c) Express the joint characteristic function of X and Y in terms of the characteristic function of X.

(d) Let $X \sim \mathsf{N}(0, 1)$ and $p = 1/2$. Show that X and Y are uncorrelated Gaussian random variables that are not independent. Comment on this result.

Answers

1. $M_{ZS}(\omega_1,\omega_2) = e^{i(\omega_1+\omega_2)m_S - \frac{1}{2}\sigma_S^2(\omega_1+\omega_2)^2} e^{-\frac{1}{2}\omega_1^2\sigma_N^2}$
2.
3. $(1-2i\omega)^{-n/2}$
4. $M_Y(\omega_1,\omega_2) = e^{-\frac{3}{2}\omega_1^2} e^{-\frac{1}{2}\omega_2^2}$, Y_1 and Y_2 are independent, $f_{Y_1}(y) = \dfrac{1}{\sqrt{6\pi}} e^{-y^2/6}$
5. $\begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix}$
6. $K_Y = \begin{pmatrix} 4 & 4 \\ 4 & 7 \end{pmatrix}$, $\quad f_{Y_1}(y) = \frac{1}{\sqrt{8\pi}} e^{-(y-1)^2/8}$, $f_{Y_2}(y) = \frac{1}{\sqrt{14\pi}} e^{-(y+1)^2/14}$, $\quad \mathbb{P}(X_3 < X_1 + X_2) = F_{N(0,1)}\left(\frac{1}{2}\right)$
7.
8. $\mathbb{E}(X) = 2$, $\mathrm{Var}(X) = 5$, $\mu_{11;XY_1} = 3$
9. $M_{SR}(\omega_1,\omega_2) = e^{-\omega_1^2-\omega_2^2}$, $\quad M_S(\omega_1) = e^{-\omega_1^2}$, $M_R(\omega_2) = e^{-\omega_2^2}$
10. $1 - F_{\mathsf{N}(0,1)}\left(\dfrac{n-1}{\sqrt{2n+2}}\right)$
11. 2
12. X_1 and X_2, X_2 and X_3, X_1 and Y.
13.
14. $\mathsf{N}(0,\sigma^2)$, where $\sigma^2 = \frac{11}{13}$.
15. (a) $(0,0)^t$, $\boldsymbol{K} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $\quad$ (b) $f_Y(y) = \frac{1}{\sqrt{4\pi}} e^{-\frac{y^2}{4}}$.
16. (c) $M_{XY}(\omega_1,\omega_2) = p\, M_X(\omega_1+\omega_2) + (1-p)\, M_X(\omega_1-\omega_2)$, $\quad$ (d) $\mathrm{Cov}(X,Y) = \mathbb{E}(X^2)\mathbb{E}(S) = 0$